MMP 1
Serie 2
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1 MC

1.a

(B)

1.b

(C)

2

$$\begin{aligned}\lim_{R\to\infty}\int_0^R \frac{\sin(x)}{x}dx &= \lim_{R\to\infty}\int_0^R\int_0^\infty e^{-tx}\sin(x)dtdx \\ &= \lim_{R\to\infty}\int_0^\infty\int_0^R e^{-tx}\sin(x)dxdt\end{aligned} \tag{2.1}$$

$$\begin{aligned}\int e^{-xt}\sin(x)dx &= -\frac{1}{t}e^{-xt}\sin(x) - \int -\frac{1}{t}e^{-xt}\cos(x)dx \\ &= -\frac{1}{t}e^{-xt}\sin(x) - \frac{1}{t^2}e^{-xt}\cos(x) - \int \frac{1}{t^2}e^{-xt}\sin(x)dx\end{aligned} \tag{2.2}$$

$$\Rightarrow \int e^{-xt}\sin(x)dx = \frac{-1}{1+\frac{1}{t^2}}\left[\frac{1}{t}e^{-xt}\sin(x) + \frac{1}{t^2}e^{-xt}\cos(x)\right] \tag{2.3}$$

$$\begin{aligned}\Rightarrow \int_0^\infty \lim_{R\to\infty}\frac{-1}{1+\frac{1}{t^2}}\left[\frac{1}{t}e^{-Rt}\sin(R) + \frac{1}{t^2}e^{-Rt}\cos(R) - \frac{1}{t^2}\right]dt &= \int_0^\infty \frac{\frac{1}{t^2}}{1+\frac{1}{t^2}}dt \\ &= \int_0^\infty \frac{1}{1+t^2}dt \\ &= \int_0^{\frac{\pi}{2}} \frac{\frac{1}{\cos^2}(x)}{\frac{\cos^2(x)+\sin^2(x)}{\cos^2(x)}}dx = \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2}\end{aligned} \tag{2.4}$$

3

3.a

Split E into pieces of max size length $\frac{1}{n}$ in one dimension.

$$T_n = \left\{c_i \subset E : \bigcup_i c_i = E, c_i \cap c_j = \varnothing, j \neq i, |a-b| < \frac{1}{n}\forall a,b \in c_i\right\} \tag{3.1}$$

Let

$$\delta(xc) = \begin{cases}0 & x \notin c \\ 1 & x \in c\end{cases} \tag{3.2}$$

define φ_n

$$\varphi_n(x) = \sum_{\sigma\in T_n}\delta_{x\sigma}\min\{f(x) : x \in \sigma\} \tag{3.3}$$

3.b

since E_j lebesque messbar

$$\mu(E_i) = \mu(E_i \cap E) + \mu(E_i^c \cap E) \tag{3.4}$$

$$\begin{aligned}\Rightarrow \lim_{n\to\infty}\int_{E_n}\varphi dx &= \lim_{n\to\infty}\left(\int_{E_n\cap E}\varphi dx + \int_{E_n^c\cap E}\varphi dx\right) \\ &= \int_E \varphi dx + \int_\varnothing \varphi dx \\ &= \int_E \varphi dx\end{aligned} \tag{3.5}$$

3.c

define $g(x)$ such that

$$\lim_{n\to\infty} \varphi_n \geq g(x), \quad \forall x \in E \tag{3.6}$$

$$\Rightarrow \lim_{n\to\infty} \int_E \varphi_n dx \geq \int_E g dx \tag{3.7}$$

3.d

define $g(x)_j$

$$g(x)_j = \begin{cases} \varphi(x) & x \in E_j \\ 0 & x \notin E_j \end{cases} \tag{3.8}$$

With (3.5) and (3.3) we get

$$\lim_{n\to\infty} \int_E \varphi_{n(x)} dx \leq \int_E f(x) dx = \int_E g dx \leq \lim_{n\to\infty} \int_E \varphi_n dx \tag{3.9}$$

Sandwiche Lemma

$$\Rightarrow \int_E f(x) dx = \lim_{n\to\infty} \int_E \varphi_n(x) dx \tag{3.10}$$

4

4.a

Sei f, g messbare funktionen auf E

$$\begin{aligned} ||fg||_1 &= \int_E f|g d|x \\ &\leq \int_E \frac{|f|^p}{p} + \frac{|g|^q}{q} dx, \qquad \forall q, p > 1 : \frac{1}{p} + \frac{1}{q} = 1 \\ &= \frac{1}{p}(||f||_p)^p + \frac{1}{q}(||g||_q)^q = 1 = ||f||_p ||g||_q \end{aligned} \tag{4.1}$$

4.b

$f, g \quad 0 < ||f||_p, ||g||_q < \infty$

$$\begin{aligned} \frac{||fg||_1}{||f||_p ||g||_q} &= \int_E \frac{|f|}{||f||_p} \frac{|g|}{||g||_p} dx \\ &\leq \int_E \frac{1}{p} \frac{f^p}{||f||_p^p} + \frac{1}{q} \frac{g^q}{||g||_q^q} dx \\ &= \frac{1}{p} + \frac{1}{q} = 1 \end{aligned} \tag{4.2}$$

$$\Rightarrow ||fg||_1 \leq ||f||_p ||g||_q \tag{4.3}$$

4.c

$$\begin{aligned} \frac{||fg||_1}{||f||_p ||g||_q} &= \int_E \frac{|fg|}{||f||_p} \frac{1}{||g||_p} dx \\ &\leq \int_E \frac{|f|}{||f||_p} \frac{|g|}{||g||_p} dx \\ &\leq \int_E \frac{1}{p} \frac{f^p}{||f||_p^p} + \frac{1}{q} \frac{g^q}{||g||_q^q} dx \\ &= \frac{1}{p} + \frac{1}{q} = 1 \end{aligned} \tag{4.4}$$

$$\Rightarrow ||fg||_1 \leq ||f||_p ||g||_q \tag{4.5}$$

4.d

$$
\begin{aligned}
||f+g||_p &= \left(\int_E |f+g|^p dx\right)^{\frac{1}{p}} \\
&\leq \left(\int_E |f||f+g|^{p-1} dx + \int_E |g||f+g|^{p-1} dx\right)^{\frac{1}{p}} \qquad (4.6)\\
&\leq \left(||f||_p \cdot ||f+g||_p^{p-1} + ||g||_p \cdot ||f+g||_p^{p-1}\right)^{\frac{1}{p}} \\
&= ||f+g||_p^{-1}\left(||f||_p + ||g||_p\right)^{\frac{1}{p}} \\
\Rightarrow \frac{||f+g||_p^p}{||f+g||_p^{p-1}} &= ||f+g||_p \leq ||f||_p + ||g||_p \qquad (4.7)
\end{aligned}
$$

5

Let $x, y \in [0,1]$. Define:

$$x \sim y \Leftrightarrow x - y \in \mathbb{Q} \tag{5.1}$$

The set V is the set of the representative of $[0,1]/\sim$

5.a

$|q| \leq 1 \in Q, v \in V \subset [0,1]$

$$
\begin{aligned}
q + v &\leq 1 + 1 = 2 \\
q + v &\geq -1 + 0 = -1
\end{aligned} \tag{5.2}
$$

$$\Rightarrow \bigcup_{\substack{q\in Q \\ |q|\leq 1}} (q+V) \subset [-1,2] \tag{5.3}$$

$\forall x \in [0,1]$

$$\exists! v \in V, \quad v \sim x \Rightarrow x - v \in \mathbb{Q} \tag{5.4}$$

Let $x - v = q \in \mathbb{Q}$

$$q + V \Rightarrow x - v + v = x \tag{5.5}$$

$$\Rightarrow [0,1] \subset \bigcup_{\substack{q\in Q \\ |q|\leq 1}} (q+V) \tag{5.6}$$

5.b

$$\mu([0,1]) = 1 \leq \mu\left(\bigcup_{\substack{q\in Q \\ |q|\leq 1}} (q+V)\right) \leq \mu([-1,2]) = 3 \tag{5.7}$$

$q \in \mathbb{Q}, v \in \mathbb{Q}, \quad \forall v \in V$

$$\Rightarrow \mu(V+q) = \mu(V) \tag{5.8}$$

$$\Rightarrow 1 \leq \mu(V) \leq 3. \tag{5.9}$$

5.c

In b) haben wir gesehen falls V Lebesque messbar dann hat es eine gewisse grösse. Aber mann kann jede $x \sim y$ als eine $q \in \mathbb{Q}$ darstellen was eine grösse 0 hat. Falls zwei Äquivalentrelation zur selben zahl in $\mathbb{Q}$ zugeordnet wird kann man das simpel mit einer Translation begründen. Somit hätte es die grösse 0 und 3 zur gelichen Zeit was nicht geht.

6

6.a

$$f_j < f, \quad \forall j \in \mathbb{N} \tag{6.1}$$

$$\Rightarrow \lim_{j\to\infty} \int_E f_j dx \leq \int_E f dx \tag{6.2}$$

6.b

Let $\varepsilon > 0$, $a \in (0,1)$ such that $1 - a \leq \varepsilon$
Then $\exists N \in \mathbb{N}$

$$|f - f_n| \leq \varepsilon, \quad \forall n \geq N \tag{6.3}$$

Since $0 \leq f$

$$\begin{aligned} \Rightarrow \int_E a\varphi dx \leq (1-\varepsilon) \int_E f dx \leq \int_E f_n dx \quad \forall n \geq N \\ \Rightarrow \int_E a\varphi dx \leq \lim_{j\to\infty} \int_E f_j dx \end{aligned} \tag{6.4}$$

6.c

In (6.4) haben wir gesehen:

$$\forall a \in (0,1), \quad \exists N \in \mathbb{N} \text{ such that } a\varphi < f_n, \quad \forall n \geq \mathbb{N} \tag{6.5}$$

Da a auf $(0,1)$ beliebig:

$$f_1 \leq f_2 \leq ... \leq f_n \leq ... = f \tag{6.6}$$

$\varepsilon \to 0$

$$\lim_{n\to\infty} \int_E f_n dx = \int_E \lim_{n\to\infty} f_n dx = \int_E f dx \tag{6.7}$$